

A Decision-Theoretic Framework for Utility-Aligned AI Explanations

Internship Proposal 2026

Topics:	Causal Inference, Decision Theory, Explainable AI
Laboratories:	INSA Rouen Normandy, LITIS & Université de Caen Normandie, GREYC
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Duration:	6 months
Level:	M2 or last year Engineering School
Application:	Send an email to the advisors with a detailed CV + academic transcripts
Obs.:	May lead to a PhD thesis

Abstract

Explanations in machine learning systems are often evaluated based on their fidelity or simplicity rather than their impact on human decision-making. This internship project aims to explore a decision-theoretical framework to construct explanations that are not only realistic but also aligned with how agents interpret and act upon information. Building on causal inference and decision theory, we will model explanations as information structures that modify beliefs and influence actions, optimizing them for expected decision utility. A prototype explainer will be developed and evaluated against standard methods such as LIME and SHAP using both quantitative metrics and simulated decision-making experiments.

1. Motivation & Problem statement

Machine learning (ML) systems are increasingly operating in high-stakes decision-making contexts, such as medical diagnosis, loan approval, and criminal justice, and their predictions typically inform human decisions directly. In such a context, there is a growing need to understand their underlying process either by employing interpretable-by-design approaches or through post-hoc explanations such as LIME [Ribeiro et al., 2016] and SHAP [Lundberg and Lee, 2017]. However, these approaches have primarily focused on descriptive fidelity (i.e., “what the model did”) rather than the decision utility of the human who uses the explanation. In other words, it is not enough that an explanation “correctly describes” the internals of a model; what matters is that the explanation improves the downstream decision-making outcome. We advocate for a causal decision-theoretic framework for AI explanations [Dawid, 2015]. It means treating an explanation as a signal that influences a human’s belief and, thereby, action aligned with a causal model (e.g., a structural causal model (SCM) [Pearl, 2009]). The objective is thus to design explanations that maximize the expected utility of the human decision, subject to cognitive/information constraints.

2. Research objective

This project aims to develop and evaluate a decision-theoretical framework for producing and assessing explanations that improve the expected utility of human decision-makers. Specifically, it will: (a) model explanations as signals that modify agents’ beliefs; (b) define realism as the consistency between an explanation, a causal model [Pearl, 2009] and the decision context; and (c) implement a prototype explainer that optimizes explanations for expected decision utility, subject to information constraints (e.g., bounded rationality, rate–distortion theory, or rational inattention). Formally, if we denote X the features, Y the outcome, $E \in \mathcal{E}$ an explanation derived from $(X, \hat{Y})$ and a domain structural causal model, and human action A . A human sees an explanation E and forms a posterior $P(Y | X, E)$. Thus, their action A is chosen to maximize utility $\mathbb{E}[U(a, Y) | X, E]$. Consequently, we optimize the explanation as follows.

$$E^* \in \arg \max_{E \in \mathcal{E}} \mathbb{E}_{X,Y} \left[\max_{a \in \mathcal{A}} \mathbb{E}[U(a, Y) | X, E] \right]$$

subject to information-capacity constraint $I(E; Y) \leq C$ or cognitive complexity $H(E) \leq H_{\max}$, and the causal validity is enforced by restricting $\mathcal{E}$ to respect an structural causal model.

3. Research questions

This objective will be pursued through the following research questions:

1. How to model an explanation E as an information channel aligned with a causal model that influences beliefs $P(Y | X, E)$ and actions A ?

2. How to encode cognitive/information constraints (e.g., limited attention, bounded rationality) on explanations?
3. Given a target utility function $U(a, y)$ for the human decision-maker, how to optimize explanation design (i.e., what features of the model to expose, what causal relations to highlight) to maximize expected utility?
4. How to empirically compare utility-aligned explanations against popular attribution baselines (e.g., LIME, SHAP) on decision utility, fidelity, stability, and causal consistency?

4. Timeline

1. Literature review on causal decision theory and rational XAI.
2. Formalize a causal decision-theoretic framework for AI explanations.
3. Implement a prototype to be integrated into an open-source framework.
4. Evaluate the prototype using 1-2 benchmark datasets (e.g., COMPAS, US Census [Ding et al., 2021], Adult Income).
5. Analyze the results and refine the framework.
6. Wrap up the findings as a scientific paper to be submitted to a workshop or conference.

4.1 Research Laboratories

- The **GREYC** (Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen, UMR 6072) is a joint research unit of the University of Caen Normandy (UNICAEN), the CNRS, and the ENSICAEN (École Nationale Supérieure d'Ingénieurs de Caen). It brings together researchers and engineers working in the fields of computer science, image and signal processing, automation, and instrumentation, with the goal of advancing both theoretical and applied research in information and communication technologies. The laboratory is organized into several specialized research groups, including those focused on artificial intelligence, machine learning, data analysis, image and text processing, cybersecurity, and embedded systems.
- The **LITIS** (Laboratoire d'Informatique, du Traitement de l'Information et des Systèmes) is a public research unit (EA 4108) in computer science, jointly affiliated with the University of Rouen Normandy, the INSA Rouen Normandy, and the University of Le Havre Normandy. A member of the CNRS research federation NormaSTIC (FR 3638), LITIS conducts advanced research in machine learning, pattern recognition, information processing, and intelligent systems. Its activities cover a broad spectrum, ranging from fundamental algorithmic research to applied system design, in domains such as health, smart mobility, and smart territories.

5. Profile

The internship requires some curiosity about causality, eXplainable AI (XAI), information theory, and the ability to work on cross-disciplinary problems, besides excellent machine learning skills and programming expertise, preferred in Python.

6. Application

Interested candidates should apply by sending: (a) a detailed CV and (b) the academic transcripts to aleite@insa-rouen.fr and nadjat.bourdache@unicaen.fr.

This internship may lead to a PhD thesis.

References

- A Philip Dawid. Statistical causality from a decision-theoretic perspective. *Annual Review of Statistics and Its Application*, 2(1):273–303, 2015.
- Frances Ding, Moritz Hardt, John Miller, and Ludwig Schmidt. Retiring adult: New datasets for fair machine learning. *Advances in neural information processing systems*, 34:6478–6490, 2021.
- Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 2017.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “why should I trust you?” explaining the predictions of any classifier. In *22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.